

James Ruse Agricultural High School
Year 12 2009 Term 1 Extension 1

Question 1.

- (a) Find A if $\sum_{k=1}^4 k \ln k = \ln A$
- (b) (i) Express $1.\dot{7}\dot{2}$ as an infinite series.
(ii) Hence write $1.\dot{7}\dot{2}$ as an equivalent fraction in simplest terms.
- (c) Find $\frac{d}{dx}(\operatorname{cosec}(x^2 + 1))$.
- (d) Find $\int \cos x e^{\sin x} dx$.
- (e) (i) Write $3\sqrt{3} \sin x - 3 \cos x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < 2\pi$.
(ii) Find the first value of $x > 0$ when $3\sqrt{3} \sin x - 3 \cos x$ is a maximum.

Question 2.

- (a) Differentiate with respect to x in simplest terms :
- (i) $\frac{\tan 3x}{\sin 3x}$.
- (ii) $e^{5x} \ln x^2$.
- (b) Find $\int \frac{9+x}{4+x^2} dx$.
- (c) (i) Simplify $\frac{9}{2x+1} + \frac{4}{3x+4}$.
(ii) Hence evaluate $\int_0^1 \frac{7x+8}{6x^2+11x+4} dx$.

Question 3.

- (a) Find the area bounded by the curve $y = \tan x$, the x axis and the line $x = \frac{\pi}{3}$.
- (b) If the sum S_n of n terms of a sequence is given by the formula $S_n = 8n^2 - 6n$, find the formula for the n th term T_n of the sequence.
- (c) Find the formula S_n for the summation of n terms of the sequence :

$$2\frac{1}{2} + 4\frac{1}{4} + 6\frac{1}{8} + 8\frac{1}{16} + \dots\dots\dots$$

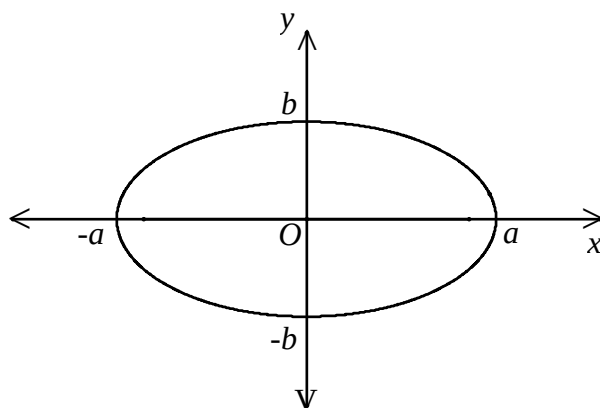
- (d) (i) Sketch the graph of $y = \frac{3x+4}{2x-1}$.

(ii) Find the values of x for a sum to infinity to exist :

$$1 + \left(\frac{3x+4}{2x-1}\right) + \left(\frac{3x+4}{2x-1}\right)^2 + \dots\dots\dots$$

Question 4.

- (a) (i) Use the substitution $x = r \sin \theta$ and evaluate $\int_0^r \sqrt{r^2 - x^2} \, dx$, where r is a constant.
- (ii) An ellipse with the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is shown below.



Using (i) find the area of the ellipse.

- (b) A man invests \$180 000 at 4% p.a. at the beginning of the month. Interest is credited at the end of each month. If the man withdraws \$ 2000 at the end of each month, find the number of \$ 2000 withdrawals that can be made.

Question 5.

- (a) Given $y = \cos(x^2)$ in the domain $0 \leq x \leq \pi$ then :
- (i) Show $\frac{d^2y}{dx^2} = -2\sin(x^2) - 4x^2 \cos(x^2)$
- (ii) Find the location of the turning points, and determine their nature.
- (iii) Graph $y = \cos(x^2)$ in the domain $0 \leq x \leq \pi$.
- (b) A helicopter which is initially on the ground, rises vertically at a constant speed of $5m/s$. If a man is initially 50 metres horizontally from the helicopter, find the rate of change of the angle of elevation θ of the helicopter when the helicopter is 20 metres above the ground.

Question 6.

- (a) Without using Calculus and using a scale of $1cm = 1$ unit on each axis, graph the following functions on the same axes in the domain $0 < x < 2\pi$:
- (i) $y = \cos x$
- (ii) $y = \cos 2x$
- (iii) $y = \cos x + \cos 2x$
- (b) Find the volume of revolution when the region bounded by the curve $y = \ln(x+1)$, The y axis and the line $y = \ln 2$ is rotated around the y axis.
- (c) Prove by Mathematical Induction :
- $$1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \dots n(n+1)^2 = \frac{n}{12}(n+1)(n+2)(3n+5)$$

End of Exam

(1)

Q1
(a) $\sum_{k=1}^4 \ln k = \ln 1 + \ln 2 + \ln 3 + \ln 4$

$$= \ln 2^2 \cdot 3 \cdot 4^2$$

$$= \ln A$$

$$A = 2^2 \cdot 3 \cdot 4^2$$

$$= 27648$$

(b) (i) $\ln 72 = 1 + \frac{72}{100} + \frac{72}{10000} + \dots$

(ii) $= 1 + \frac{\frac{72}{100}}{1 - \frac{1}{100}}$

$$= 1 + \frac{72}{99}$$

$$= 1 \frac{8}{11}$$

(c) $\frac{d}{dx} (\sec(x^2)) = 2x \sec(x^2) \tan(x^2)$

(d) $\int \cos x e^{\sin x} dx = e^{\sin x} + C$

(e) (i) $R \cos(x-d) = R \cos x \cos d + R \sin x \sin d$
 $R \sin d = 3\sqrt{3}$ $R \cos d = -3$

$$R = \sqrt{(3\sqrt{3})^2 + 3^2}$$

$$= \sqrt{36}$$

$$= 6$$

$$\tan d = -\sqrt{3}$$

$$d = \frac{2\pi}{3}$$

$$\therefore 3\sqrt{3} \sin x - 3 \cos x = 6 \left(\cos \left(x - \frac{2\pi}{3} \right) \right)$$

(ii) $x - \frac{2\pi}{3} = 0$

$$x = \frac{2\pi}{3}$$

Q2(a) (i) $\frac{d}{dx} \left(\frac{\tan 3x}{\sin 3x} \right) = \frac{d}{dx} (\sec 3x)$

$$= 3 \sec 3x \tan 3x$$

(ii) $\frac{d}{dx} e^{5x} \ln x^2 = \frac{d}{dx} (2e^{5x} \ln x)$

$$= 2 \cdot \frac{e^{5x}}{x} + 10e^{5x} \ln x$$

$$= 2e^{5x} \left[\frac{1}{x} + 5 \ln x \right]$$

(b) $\int \frac{9+x}{4+x^2} dx = \int \left(\frac{9}{4+x^2} + \frac{x}{4+x^2} \right) dx$
 $= \frac{9}{2} \tan^{-1} \frac{x}{2} + \frac{1}{2} \ln(x^2+4) + C$

(c) (i) $\frac{9}{2x+1} + \frac{4}{3x+4} = \frac{9(3x+4) + 4(2x+1)}{(2x+1)(3x+4)}$
 $= \frac{35x + 40}{6x^2 + 11x + 4}$

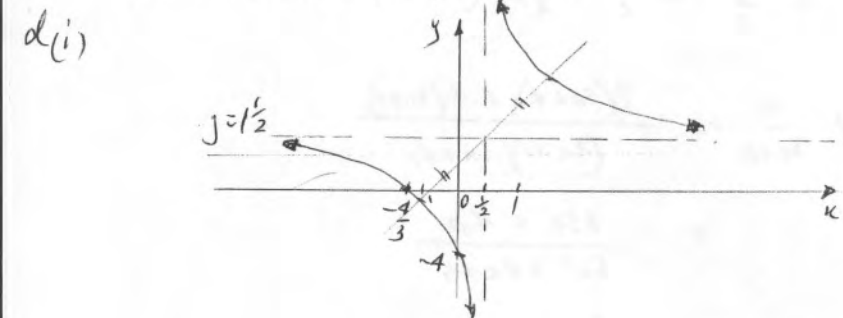
(ii) $\int_0^1 \frac{7x+8}{6x^2+11x+4} dx = \frac{1}{5} \int_0^1 \left(\frac{9}{2x+1} + \frac{4}{3x+4} \right) dx$
 $= \frac{1}{5} \left[\frac{9}{2} \ln(2x+1) + \frac{4}{3} \ln(3x+4) \right]_0^1$ as for $x > 0$
 $= \frac{1}{5} \left[\frac{9}{2} \ln\left(\frac{3}{1}\right) + \frac{4}{3} \ln\left(\frac{7}{4}\right) \right]$

Q3(c) Area = $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{\sin x}{\cos x} dx$
 $= \left[-\ln |\cos x| \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $\cos x > 0$ for $0 < x < \frac{\pi}{2}$
 $= - \left[\ln \frac{1}{2} - 0 \right] = \ln 2$ square units

(2)

$$\begin{aligned} \text{a)} \quad S_n &= 8n^2 - 6n \\ S_{n-1} &= 8(n-1)^2 - 6(n-1) \\ T_n &= S_n - S_{n-1} \\ &= 16n - 14 \end{aligned}$$

$$\begin{aligned} \text{c)} \quad S_n &= 2\frac{1}{2} + 4\frac{1}{4} + 6\frac{1}{8} + 8\frac{1}{16} \\ &= (2 + 4 + \dots + 2n) + \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^n}\right) \\ &= \frac{n}{2} [4 + (n-1) \cdot 2] + \frac{1}{2} \left(\frac{1 - (\frac{1}{2})^n}{1 - \frac{1}{2}} \right) \\ &= n(n+1) \times \frac{1}{2} + 1 - \left(\frac{1}{2}\right)^n \\ S_n &= n^2 + n + 1 - \left(\frac{1}{2}\right)^n \end{aligned}$$



$$\begin{aligned} \text{(ii)} \quad \text{For limiting sum} \quad -1 < r < 1 \quad r \neq 0. \\ \frac{3x+4}{2x-1} &= 1 & \frac{3x+4}{2x-1} &= -1 \\ x &= -5 & x &= -\frac{3}{5} \end{aligned}$$

$$\therefore \text{Soln} \quad \left\{ -5 < x < -\frac{4}{3} \right\} \text{ OR } \left\{ -\frac{4}{3} < x < -\frac{3}{5} \right\}$$

$$\begin{aligned} \text{(3)} \quad \text{Q4(i)} \quad \int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2 \sin^2 \theta} \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \sqrt{r^2 (1 - \sin^2 \theta)} \cdot r \cos \theta \, d\theta \\ &= \int_0^{\frac{\pi}{2}} \sqrt{r^2 \cos^2 \theta} \cdot r \cos \theta \, d\theta \\ &= r^2 \int_0^{\frac{\pi}{2}} \cos^2 \theta \, d\theta \\ &= r^2 \int_0^{\frac{\pi}{2}} \frac{1 + \cos 2\theta}{2} \, d\theta \\ &= \frac{r^2}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{2}} \\ &= \frac{r^2}{2} \left[\frac{\pi}{2} + 0 - 0 \right] \\ &= \frac{\pi r^2}{4} \end{aligned}$$

(1)

$$\begin{aligned} \text{(ii)} \quad \text{Area} \\ \text{ellipse} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \\ y = b \sqrt{1 - \frac{x^2}{a^2}} \\ = \frac{b}{a} \sqrt{a^2 - x^2} \\ \text{Area} = 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} \, dx \\ = 4 \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} \, dx \\ = 4 \frac{b}{a} \cdot \frac{\pi a^2}{4} = \pi ab \end{aligned}$$

(1)

Q5
(a)(i)

$$y = \cos x^2$$

$$\frac{dy}{dx} = -2x \sin x^2$$

$$\frac{d^2y}{dx^2} = -2x \cdot \cos x^2 \cdot 2x - 2 \sin x^2$$

$$= -4x^2 \cos x^2 - 2 \sin x^2$$

(ii) For turning points

$$\frac{dy}{dx} = 0$$

$$-2x \sin x^2 = 0$$

$$x = 0 \quad x^2 = 0, \pi, 2\pi, 3\pi, \dots$$

$$x = 0, \pm\sqrt{\pi}, \pm\sqrt{2\pi}, \pm\sqrt{3\pi}, \dots$$

But $x = \sqrt{\pi}, \sqrt{2\pi}, \sqrt{3\pi}$ as $0 < x < \pi$
 $y = -1 \quad y = 1 \quad y = 1$

Turning points $(\sqrt{\pi}, -1), (\sqrt{2\pi}, 1)$ & $(\sqrt{3\pi}, -1)$ ①

For nature of turning points test $\frac{d^2y}{dx^2}$ for sign concavity
 as $0 < x < \pi$ is continuous.

At $x = \sqrt{\pi} \quad \frac{d^2y}{dx^2} = -2 \cdot 0 - 4\pi \cdot -1$
 $\frac{d^2y}{dx^2} = 4\pi$

> 0
 $\therefore$ Relative minimum at $(\sqrt{\pi}, -1)$

At $x = \sqrt{2\pi} \quad \frac{d^2y}{dx^2} = -2 \cdot 0 - 4 \cdot 2\pi \cdot 1$
 $\frac{d^2y}{dx^2} = -8\pi$

< 0
 $\therefore$ Relative maximum at $(\sqrt{2\pi}, 1)$

At $x = \sqrt{3\pi} \quad \frac{d^2y}{dx^2} = -2 \cdot 0 - 4 \cdot 3\pi \cdot -1$
 $\frac{d^2y}{dx^2} = 12\pi$

> 0
 $\therefore$ Relative minimum at $(\sqrt{3\pi}, -1)$

5

②

(b) Amount left end 1st month = $(18000)(1+r) - 2000$ monthly interest = $\frac{2}{3}\%$
 $r = \frac{1}{300}$ ⑥

Amount left end 2nd month = $[18000(1+r) - 2000](1+r) - 2000$
 $= 18000(1+r)^2 - 2000[1 + (1+r)]$

Amount left end 3rd month = $[18000(1+r)^2 - 2000(1 + (1+r))](1+r) - 2000$
 $= (18000)(1+r)^3 - 2000[1 + (1+r) + (1+r)^2]$

Amount left end n months = $18000 \cdot (1+r)^n - 2000[1 + (1+r) + \dots + (1+r)^{n-1}]$ ①
 $= 18000(1+r)^n - 2000 \left[\frac{(1+r)^n - 1}{1+r - 1} \right]$
 $= 18000(1+r)^n - 2000 \left[\frac{(1+r)^n - 1}{r} \right]$ ①

$$0 = 18000(1+r)^n - 2000 \left[\frac{(1+r)^n - 1}{r} \right]$$

$$\therefore 18000 \left(1 + \frac{1}{300}\right)^n = 2000(300) \left[\left(1 + \frac{1}{300}\right)^n - 1 \right]$$
 ①

$$420000 \left(\frac{301}{300}\right)^n = 600000$$

$$\left(\frac{301}{300}\right)^n = \frac{10}{7}$$

$$n = 107.18$$

$\therefore$ Number of \$2000 payments is 107

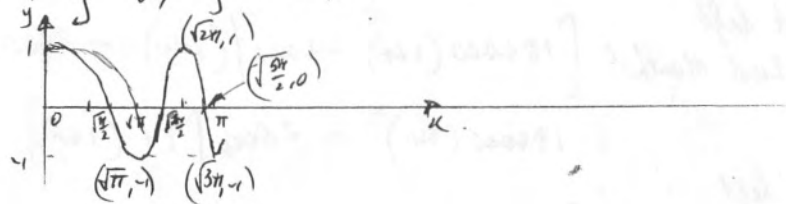
①

①

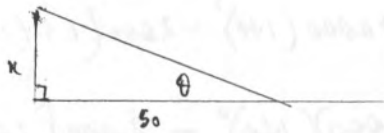
③

(iii) $x=0$ $y=1$ $y'=0$

$x=\pi$ $y=-0.9$ $y'=-2.70$



(b)



$$\tan \theta = \frac{x}{50}$$

$$\theta = \tan^{-1} \frac{x}{50}$$

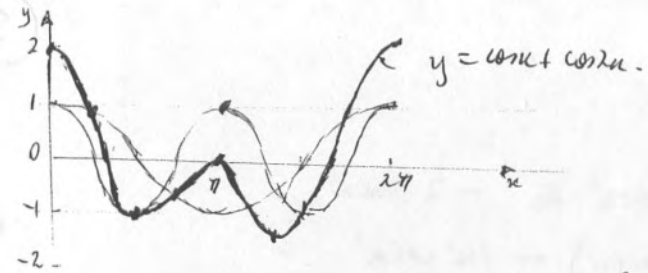
$$\frac{d\theta}{dx} = \frac{50}{2500 + x^2}$$

$$\frac{dx}{dt} = 5$$

$$\begin{aligned} \therefore \text{Rate of change angle elevation} &= \frac{d\theta}{dt} \\ &= \frac{d\theta}{dx} \cdot \frac{dx}{dt} \\ &= \frac{50}{2500 + 20^2} \cdot 5 \text{ when } x=20 \\ &= \frac{250}{2900} \\ &= \frac{5}{58} \text{ rad/s.} \end{aligned}$$

(7)

(e)



(1) each graph.

(b)

$$y = \ln(x+1)$$

$$x = e^y - 1$$

$$\begin{aligned} \text{Volume } V &= \pi \int_a^b x^2 dy \\ &= \pi \int_0^{\ln 2} (e^y - 1)^2 dy \\ &= \pi \int_0^{\ln 2} (e^{2y} - 2e^y + 1) dy \\ &= \pi \left[\frac{e^{2y}}{2} - 2e^y + y \right]_0^{\ln 2} \\ &= \pi \left[\frac{1}{2} - 4 + \ln 2 - \left(\frac{1}{2} - 2 + 0 \right) \right] \\ &= \pi \left[\ln 2 - \frac{1}{2} \right] \text{ unit}^3. \end{aligned}$$

(9)

(a)

Step 1

 $n=1$

$$T_1 = 1 \cdot (1+1)^2 = 4$$

$$S_1 = \frac{1}{12} (1+1)(1+2)(3+5)$$

$$= \frac{2 \cdot 3 \cdot 8}{12}$$

$$= \frac{48}{12}$$

$$= 4$$

$$T_1 = S_1$$

i.e. statement true $n=1$ Step 2 Assume statement is true $n=k$

$$i.e. 1 \times 2^2 + 2 \times 3^2 + \dots$$

$$k(k+1)^2 = \frac{1}{12} (k+1)(k+2)(3k+5)$$

To prove statement is true $n=k+1$

$$i.e. 1 \times 2^2 + 2 \times 3^2 + \dots$$

$$k(k+1)^2 + (k+1)(k+2)^2 = \frac{(k+1)(k+2)(k+3)(3k+8)}{12}$$

$$\text{Now } 1 \times 2^2 + 2 \times 3^2 + \dots k(k+1)^2 + (k+1)(k+2)^2$$

$$= \frac{k}{12} (k+1)(k+2)(3k+5) + (k+1)(k+2)^2$$

By assumption

$$= \frac{(k+1)(k+2)}{12} [3k^2 + 5k + 12(k+2)]$$

$$= \frac{(k+1)(k+2)}{12} [3k^2 + 17k + 24]$$

$$= \frac{(k+1)(k+2)}{12} [3k^2 + 8k + 9k + 24]$$

$$= \frac{(k+1)(k+2)}{12} [k[3k+8] + 3[3k+8]]$$

$$= \frac{(k+1)(k+2)(k+3)(3k+8)}{12}$$

$\therefore$ If statement true $n=k$ it is also true $n=k+1$
 Since statement is true $n=1$ it is also true $n=1+1=2, 2+1=3$